

WEEK – 2

ENTITY RELATIONSHIP ANALYSIS REPORT

Project Title

**ShopNest – E-Commerce Order
Management Database System**

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1. Introduction

The **ShopNest – E-Commerce Order Management Database System** uses a relational database to organize and manage business data efficiently. After identifying the entities and their attributes, the next step is to analyze how these entities are related to each other.

Relationship analysis identifies the connections between entities using **Primary Keys (PK)** and **Foreign Keys (FK)**. It also defines the **cardinality** of each relationship, such as **One-to-One (1:1)**, **One-to-Many (1:M)**, and **Many-to-Many (M:N)**. Proper relationship analysis ensures data consistency, reduces redundancy, and supports efficient database operations.

2. Relationship Analysis

Customer → Order

A customer can place multiple orders, but each order belongs to only one customer.

Relationship: Customer places Orders.

Primary Key: Customer.Customer_ID

Foreign Key: Order.Customer_ID

Category → Product

Each category contains multiple products, while each product belongs to only one category.

Relationship: Category contains Products.

Primary Key: Category.Category_ID

Foreign Key: Product.Category_ID

Order → Order Details

One order may contain multiple products. Each order detail belongs to one order.

Relationship: Order contains Order Details.

Primary Key: Order.Order_ID

Foreign Key: Order Details.Order_ID

Product → Order Details

A product can appear in many different customer orders. Each order detail refers to one product.

Relationship: Product appears in Order Details.

Primary Key: Product.Product_ID

Foreign Key: Order Details.Product_ID

Order → Payment

Each order has one payment record, and each payment belongs to one order.

Relationship: Order has Payment.

Primary Key: Order.Order_ID

Foreign Key: Payment.Order_ID

Order → Shipment

Each order has one shipment record, and each shipment belongs to one order.

Relationship: Order has Shipment.

Primary Key: Order.Order_ID

Foreign Key: Shipment.Order_ID

Customer → Review

A customer can submit multiple reviews for different products. Each review belongs to one customer.

Relationship: Customer writes Reviews.

Primary Key: Customer.Customer_ID

Foreign Key: Review.Customer_ID

Product → Review

A product can receive many customer reviews, while each review refers to only one product.

Relationship: Product receives Reviews.

Primary Key: Product.Product_ID

Foreign Key: Review.Product_ID

3. Cardinality Analysis

Relationship	Cardinality	Description
Customer → Order	One-to-Many (1:M)	One customer can place many orders.
Category → Product	One-to-Many (1:M)	One category contains many products.
Order → Order Details	One-to-Many (1:M)	One order contains many order items.
Product → Order Details	One-to-Many (1:M)	One product can appear in many order items.
Order → Payment	One-to-One (1:1)	Each order has one payment record.
Order → Shipment	One-to-One (1:1)	Each order has one shipment record.
Customer → Review	One-to-Many (1:M)	One customer can write many reviews.
Product → Review	One-to-Many (1:M)	One product can receive many reviews.

Many-to-Many Relationship

There is one indirect **Many-to-Many (M:N)** relationship in the system.

Customer $\leftrightarrow$ Product

- A customer can purchase many products.
- A product can be purchased by many customers.

This many-to-many relationship is resolved through the **Order Details** entity.

Relationship Path:

Customer $\rightarrow$ Order $\rightarrow$ Order Details $\rightarrow$ Product

4. Relationship Summary

Parent Entity	Child Entity	Relationship	Cardinality
Customer	Order	Places	One-to-Many (1:M)
Category	Product	Contains	One-to-Many (1:M)
Order	Order Details	Contains	One-to-Many (1:M)
Product	Order Details	Appears In	One-to-Many (1:M)
Order	Payment	Has	One-to-One (1:1)
Order	Shipment	Has	One-to-One (1:1)
Customer	Review	Writes	One-to-Many (1:M)
Product	Review	Receives	One-to-Many (1:M)

5. Conclusion

The **Entity Relationship Analysis** defines how the entities in the **ShopNest – E-Commerce Order Management Database System** interact with one another. The identified relationships and their cardinalities provide a clear structure for designing the Entity Relationship (ER) Diagram and implementing the relational database.

Using **Primary Keys** and **Foreign Keys** ensures referential integrity and maintains consistency across the database. This relationship analysis serves as the foundation for the next stages of the project, including **ER Diagram creation, normalization, relational schema design, and SQL implementation.**